Alternative Assessment Tool (AAT) Report on

"Design and Verification of Digital circuit using Cadence Tool"

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)

[Course Code: 23ECSPE1AD]

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1. Divide-by-N Counter

A Divide-by-N Counter is a sequential digital circuit used to divide the input clock frequency by a specified value. It counts the incoming clock pulses and changes its output after reaching a predefined count.

It is commonly used in digital systems for clock frequency division, timing circuits, and generating lower-frequency clock signals.

The counter increments its value on every active clock edge. When the counter reaches the specified terminal count (N-1), it resets to zero or changes its output state, depending on the design.

This process repeats continuously to generate a divided-frequency output signal.

Design Code:

```
module divide_n_counter #(parameter N = 10)
(
   input clk,
   input rst,
   output reg [$clog2(N)-1:0] count,
   output reg out
 );

 always @(posedge clk or posedge rst)
 begin
     if (rst)
     begin
          count <= 0;
          out <= 0;
     end
     else
     begin
          if (count == N-1)
          begin
               count <= 0;
               out <= ~out;
          end
          else
          begin
                count <= count + 1;
          end
     end
 end

 endmodule
```

Testbench Code:

```
module tb_counter;
reg clk, rst;
wire [3:0] count;
wire out;

divide_n_counter #(.N(10)) uut (
    .clk(clk),
    .rst(rst),
    .count(count),
    .out(out)
);

always #5 clk = ~clk;

initial
begin
    $shm_open("waves.shm");
    $shm_probe(tb_counter, "AS");

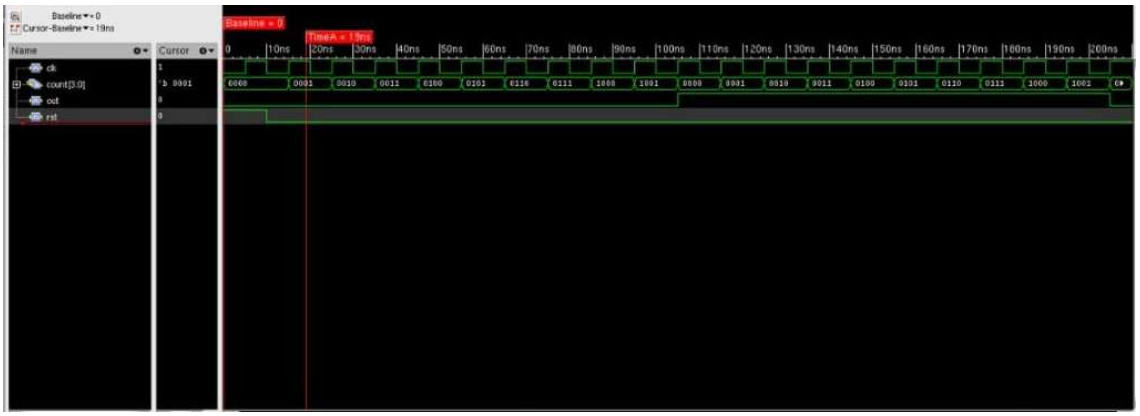
    clk = 0;
    rst = 1;

    #10 rst = 0;

    #200 $finish;
end

endmodule
```

Output Waveform:



2.COMPARATOR

A magnitude comparator is a combinational logic circuit used to compare two binary numbers and determine their relative magnitudes.

It determines whether:

- A is greater than B (A > B).
- A is equal to B (A = B).
- A is less than B (A < B).

Magnitude comparators are widely used in digital systems, processors, arithmetic circuits, and control applications.

Boolean Expressions

For a 4-bit comparator:

Equality output:

$$E = (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)$$

Greater-than output:

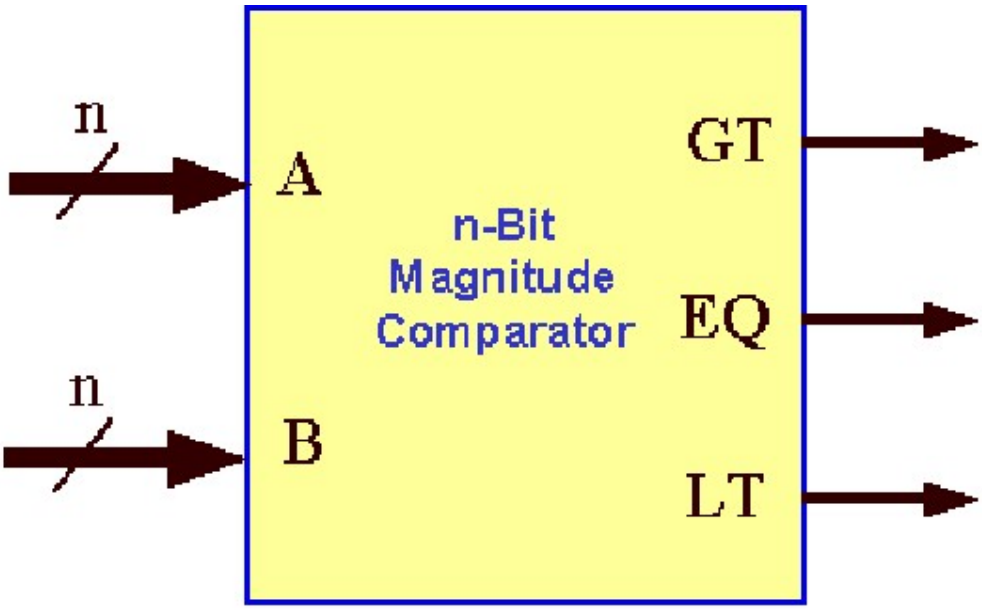
$$\begin{aligned} G = \quad & A_3\bar{B}_3 \\ & +X_3A_2\bar{B}_2 \\ & +X_3X_2A_1\bar{B}_1 \\ & +X_3X_2X_1A_0\bar{B}_0 \end{aligned}$$

Less-than output:

$$\begin{aligned} L = \quad & \bar{A}_3B_3 \\ & +X_3\bar{A}_2B_2 \\ & +X_3X_2\bar{A}_1B_1 \\ & +X_3X_2X_1\bar{A}_0B_0 \end{aligned}$$

Where $X_i = A_i \odot B_i$, and $\odot$represents XNOR.

Block Diagrams:



Design Code:

```
module comp_n #(
 parameter N=4
)( input [N-1:0]a,b,
output reg agb,alb,aeb
);
always@(*)begin
if(a>b)
begin
agb=1;
aeb=0;
alb=0;
end
else if(a<b)
begin
agb=0;
aeb=0;
alb=1;
end
else
begin
agb=0;
aeb=1;
alb=0;
end
end
endmodule
```

Testbench Code:

```
module tb_counter;
reg clk, rst;
wire [3:0] count;
wire out;

divide_n_counter #(.N(10)) uut (
    .clk(clk),
    .rst(rst),
    .count(count),
    .out(out)
);

always #5 clk = ~clk;

initial
begin
    $shm_open("waves.shm");
    $shm_probe(tb_counter, "AS");

    clk = 0;
    rst = 1;

    #10 rst = 0;

    #200 $finish;
end

endmodule
```

Output Waveform:

